

Three Dimensional Room Acoustics Models and Optimising for Performance on the GPU

Jacob Josiah Webber

Project Preparation Presentation

Introduction

This talk aims to answer the following questions:

- ▶ What is sound?
- ▶ How do the spaces we hear them in affect this sound?
- ▶ What methods currently exist to simulate this?
- ▶ Why would we want to simulate this?
- ▶ How will I try to improve these?

Sound

What is it?

From Wikipedia

“In physics, sound is a vibration that propagates as a typically audible mechanical wave of pressure and displacement, through a transmission medium such as air or water.”

From my undergraduate degree

$$\nabla^2 \mathbf{p} - \frac{1}{c^2} \frac{\partial^2 \mathbf{p}}{\partial t^2} = 0 \quad (1)$$

Room acoustics

- ▶ The physical properties of rooms affects how we hear sound.
- ▶ Sound reverberates off walls/objects.
- ▶ Waves diffract around corners.
- ▶ Some rooms echo, some sound dry.
- ▶ Backwards: If we know physical properties can we derive the sound?

The wave equation

$$\frac{\partial^2 \mathbf{p}}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 \mathbf{p}}{\partial t^2} = 0$$

where $\mathbf{p}$ is pressure deviation and c is speed of sound in medium.

- ▶ Very familiar to those who have studied physics
- ▶ It, and its application to acoustics, are derivable from first principles.
- ▶ Solving for $\mathbf{p}$ gives us acoustic properties of a system (boundary conditions are very important!).
- ▶ Solving this analytically for complex systems is not possible.

FTDT

Finite-difference time-domain method

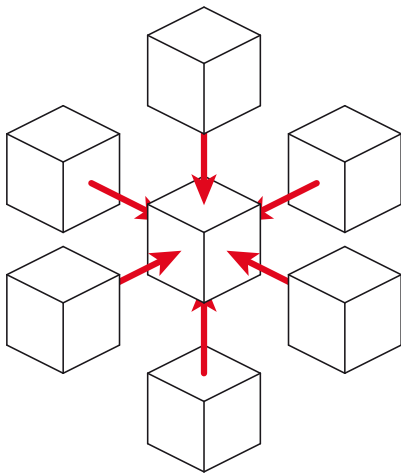
Finite difference methods are a way of approximating and discretising differential equations so that they can be solved numerically.

Effectively turns solving differential equation into a large stencilling operation.

Simple 1 dimensional example:

$$\frac{df(x)}{dx} \approx \frac{f(x+h) - f(x-h)}{2h}$$

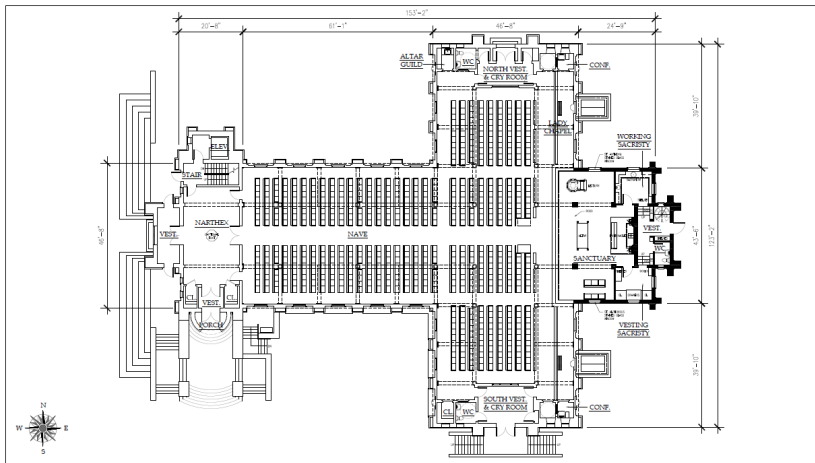
Stencil



Memory considerations

For stencilling operations, memory access is the bottleneck.

- ▶ Accurate models need a lot of points to model a room.
- ▶ Wavelength of sound at 20kHz is 17mm.
- ▶ High memory bandwidth means GPUs are used - limited memory.
- ▶ Standard 3d arrays are cuboid.
- ▶ Due to memory layout, neighbouring points in space will always be far apart in memory.



McCREERY Architects
100 Main Street, Suite 100
Rumson, NJ 07070
(201) 717-1144 www.mccreeryarchitects.com

HOLY CROSS CHURCH RUMSON, NEW JERSEY

Main Floor Plan
1/16" = 1'-0"
December 29, 2011
Sheet 4 of 21

The Proposal

Use an array of structs instead!

- ▶ Store enough points for CUDA kernel to operate on.
- ▶ Only store structs with at least one point inside the room.
- ▶ Experiment with different sized arrays within the struct to see if caching benefits can be yielded.

The Struct

Listing 1: First 2d example

```
struct linked_struct {  
    int up;  
    int down;  
    int left;  
    int right;  
    float value[N][M];  
    float old_value[N][M];  
    char inside[N][M];  
};
```

The plan

What will I do?

- ▶ Implement struct in 2d, then 3d
- ▶ Implement algorithm to convert from bounded array to array of structs.
- ▶ See if this uses less memory (it does!)
- ▶ See if this method is intolerably slow in CUDA.
- ▶ See if results are **correct**.
- ▶ Experiment with different sub-array sizes to optimise for
 - ▶ memory usage
 - ▶ caching efficiency

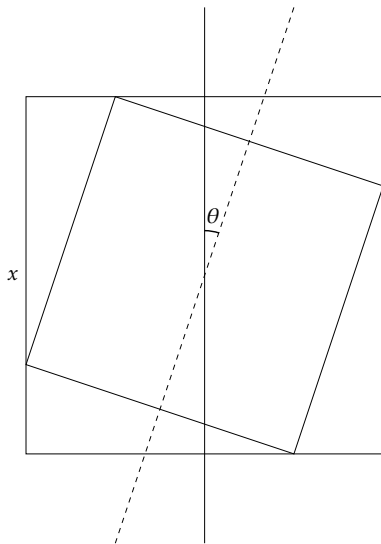


Figure : Diagram showing unit square rotated by θ within bounding square, where x is the side of the bounding square.

The End

This talk aimed to answer the following questions:

- ▶ What is sound?
- ▶ How do the spaces we hear them in affect this sound?
- ▶ What methods currently exist to simulate this?
- ▶ Why would we want to simulate this?
- ▶ How will I try to improve these?